AN INTEGRATED AI AND IoT-BASED FRAMEWORK FOR AUTOMATED CEMENT QUALITY ASSESSMENT

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ABSTRACT

The cement industry requires accurate and efficient quality control methods to ensure structural safety and material performance. Traditional approaches to cement quality assessment are largely manual, fragmented, and subjective, leading to inconsistencies, delayed feedback, and reduced responsiveness in industrial quality assurance workflows. This dissertation presents an integrated AI- and IoT-based framework for automated cement quality assessment, combining four principal components into a unified, web-deployable platform.

The first component employs a YOLOv9-based instance segmentation model to perform automated color-based particle identification in clinker microscopy images, classifying particles into dark red (iron-rich), light red, and white (quartz) categories from a dataset of approximately 300 microscopic images collected at INSEE Cement (Private) Limited, Sri Lanka. The second component integrates IoT-based ultrasonic sensors for non-contact cement cube dimension measurement with a U-Net deep learning architecture for real-time crack detection and compressive strength calculation, replacing error-prone manual vernier caliper measurements and subjective visual inspection.

The third component develops a hybrid ensemble machine learning framework combining XGBoost and LightGBM for multi-output cement compressive strength prediction at five curing ages (1, 2, 7, 28, and 56 days), incorporating chemical composition, physical properties, and environmental factors. The fourth component employs MobileNetV2 with transfer learning to automatically classify four cement clinker phases (C2S, C3S, C3A, and C4AF) from microscopic images.

Validation using industrial datasets from INSEE Cement achieved 88% mean Average Precision for particle identification, 90% validation accuracy for crack detection, an ensemble R2 score of 0.89 for multi-age strength prediction, and 91.6% validation accuracy for clinker phase classification. The proposed integrated framework demonstrates that AI and IoT technologies can substantially improve the consistency, speed, and reliability of cement quality assessment, offering a scalable, operator-independent solution suitable for industrial deployment.

Keywords: Cement quality control, clinker microscopy, crack detection, compressive strength prediction, YOLOv9, U-Net, MobileNetV2, XGBoost, LightGBM, industrial automation, IoT, machine learning.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ASTM	American Society for Testing and Materials
C2S	Dicalcium Silicate (Belite)
C3A	Tricalcium Aluminate
C3S	Tricalcium Silicate (Alite)
C4AF	Tetracalcium Aluminoferrite (Brownmillerite)
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DB	Database
EN	European Standard
FastAPI	Fast Application Programming Interface (Python web framework)
GELAN	Generalized Efficient Layer Aggregation Network
IDE	Integrated Development Environment
IoT	Internet of Things
ISO	International Organization for Standardization
LightGBM	Light Gradient Boosting Machine
LIMS	Laboratory Information Management System
MAE	Mean Absolute Error
mAP	Mean Average Precision
ML	Machine Learning
MobileNetV2	Mobile Neural Network Version 2
MPa	Megapascal
NDT	Non-Destructive Testing
OPC	Ordinary Portland Cement
PDF	Portable Document Format
PGI	Programmable Gradient Information
R^2	Coefficient of Determination

Abbreviation	Description
React.js	React JavaScript Library (frontend framework)
RMSE	Root Mean Squared Error
SCM	Supplementary Cementitious Material
SDLC	Software Development Life Cycle
SEM	Scanning Electron Microscopy
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
U-Net	Encoder-Decoder Convolutional Network for Image Segmentation
UI	User Interface
USB	Universal Serial Bus
UX	User Experience
VGG16	Visual Geometry Group Network (16 layers)
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once (real-time object detection framework)
YOLOv9	You Only Look Once Version 9

CHAPTER 1: INTRODUCTION

1.1 Background and Literature Survey

Cement is one of the most critical materials in modern construction, with global production exceeding 4 billion tons annually [1]. Quality control in cement manufacturing and concrete testing plays a vital role in ensuring structural safety, durability, and compliance with international standards such as ASTM, EN, and ISO specifications [2]. Even minor variations in chemical composition, fineness, or curing conditions can significantly affect the mechanical performance and long-term behavior of cement-based structures [3].

Despite its importance, traditional cement quality assessment methods face significant challenges related to efficiency, accuracy, and reliability. Microscopic examination of cement phases and clinker particles remains a fundamental practice in quality control; however, it is difficult to standardize across operators and requires a high level of expertise. Manual analysis involves visual inspection using optical microscopes, subjective interpretation of particle colors and phase boundaries, and time-consuming calculations, all of which are prone to human error and inconsistency [4]. Furthermore, these procedures are labor-intensive and unsuitable for high-throughput industrial environments.

1.1.1 Microscopy in Cement Quality Control

Machine learning and computer vision have increasingly found practical applications throughout various stages of cement production. Scanning Electron Microscopy (SEM), especially when combined with X-ray microanalysis, has proven extremely useful for precisely measuring the composition of clinker minerals such as alite (C3S), belite (C2S), ferrites (C4AF), and aluminate (C3A). This method allows for direct measurement of surface areas and phase proportions in clinker, often providing more detailed and accurate information than traditional chemical calculations such as Bogue's equations [5].

Although SEM provides high accuracy, optical microscopy using reflected or transmitted light is a more accessible and cost-effective tool. It is particularly helpful for quickly identifying mineral phases in raw meal or clinker samples by revealing important features like color differences, particle size, and shapes. Iron-rich particles typically have reddish hues, while quartz particles appear white or translucent. Optical microscopy also helps researchers understand how certain components affect reactions in the kiln, such as quartz influencing belite formation, which in turn impacts early strength development and kiln process efficiency [6].

Traditional quality control involves manually examining raw meal particles under a microscope, classifying them by visual features such as color and texture. This manual approach is laborious, slow, and prone to human error, which becomes especially problematic when handling large numbers of samples. Modern microscopy methods like SEM combined with X-ray microanalysis offer very precise clinker analysis, but these techniques are expensive and not practical for routine quality checks. Optical microscopy is more affordable but still depends heavily on human interpretation, limiting how quickly and consistently large volumes of samples can be evaluated [7].

1.1.2 Compressive Strength Testing and Crack Detection

Compressive strength testing of cement and concrete cubes remains a time-dependent process, typically requiring 7-day and 28-day curing periods before reliable results can be obtained. This delay restricts early decision-making in production control and increases the risk of material wastage and process inefficiencies [8]. Early-age strength prediction and automated crack detection have therefore gained increasing attention as effective solutions for reducing testing time while maintaining accuracy.

At present, cement cube testing relies heavily on vernier calipers to measure cube dimensions and on visual judgment to identify the first crack under compressive loading. These methods are labor-intensive, prone to human error, and provide only limited real-time feedback. Recent studies demonstrate the potential of machine learning and computer vision for analyzing cube surfaces and predicting compressive strength. Digital image processing combined with artificial neural networks has been shown to achieve high accuracy in estimating concrete strength from surface images [9], while CNN-based models such as InceptionV3 have proven highly effective in detecting cracks in laboratory-scale concrete cubes with accuracies exceeding 97% [10].

The integration of IoT sensors, particularly ultrasonic and infrared distance sensors, allows for non-contact measurement of cube dimensions, eliminating the limitations of manual calipers. However, infrared sensors are sensitive to environmental conditions and object properties, making ultrasonic sensors a more stable and accurate option for industrial applications [11]. The combination of load cell force measurement with real-time image-based crack detection represents a significant advance over conventional destructive testing methods.

1.1.3 Machine Learning for Cement Strength Prediction

Cement compressive strength is a critical parameter in the construction industry, directly influencing the durability and safety of infrastructure projects. Traditional methods for predicting cement strength rely on laboratory testing, which is time-consuming and may delay quality control decisions. Recent research has focused on leveraging machine learning and statistical techniques to predict cement strength more efficiently and accurately.

Machine learning models outperform traditional statistical methods in strength prediction. Studies show that models using chemical composition parameters (CaO, SiO2, Al2O3, Fe2O3) and physical parameters (Blaine fineness, particle size distribution) can predict cement strength with high accuracy [12]. The integration of multiple input features and advanced algorithms enables real-time prediction. However, most existing studies are based on international datasets and do not address local production conditions or real-time application in Sri Lankan cement plants [13].

Environmental factors such as curing temperature and humidity, which can significantly influence strength outcomes, are often overlooked or inadequately incorporated in existing models. Additionally, there is an absence of platforms validated with local operational data that can deliver rapid, batch-specific strength estimates at multiple curing ages. Without such tools, Sri Lankan manufacturers must rely on time-consuming laboratory tests, resulting in delays, inefficiencies, and increased risk of non-compliance with quality standards.

1.1.4 Clinker Phase Classification

The composition and distribution of clinker phases strongly influence cement hydration behavior and mechanical performance. The four primary clinker phases are Tricalcium Silicate (C3S or Alite), Dicalcium Silicate (C2S or Belite), Tricalcium Aluminate (C3A), and Tetracalcium Aluminoferrite (C4AF or Brownmillerite). Deep learning-based image classification models have demonstrated strong capability in identifying primary clinker phases directly from microscopic images, providing a reliable alternative to manual microscopic analysis [14].

Recent advancements in transfer learning, particularly using pre-trained architectures such as MobileNetV2, VGG16, ResNet50, and InceptionV3, have dramatically improved classification accuracy for materials science applications. These automated approaches improve repeatability, reduce analysis time, and support real-time industrial quality monitoring systems. However, most existing research addresses these challenges in isolation, without offering an integrated, end-to-end solution suitable for industrial deployment.

1.2 Research Gap

Despite significant advancements in cement quality assessment using machine learning, computer vision, and IoT technologies, a substantial gap remains in the practical application of these technologies within the cement industry, particularly in Sri Lanka. The key gaps identified through a comprehensive review of existing literature are summarized in Table 1.1.

Table 1.1: Comparison of Existing Research with Proposed System

Feature / Capability	Research A	Research B	Research C	Research D	Research E	Proposed System
Raw material particle analysis	No	Yes	Yes	Yes	Partial	Yes
Color-based particle segmentation	No	No	No	No	No	Yes

Feature / Capability	Research A	Research B	Research C	Research D	Research E	Proposed System
Automated ML-based approach	No	Partial	Yes	No	Yes	Yes
IoT-based non-contact measurement	No	No	Partial	No	No	Yes
Real-time crack detection	No	No	Yes	Yes	Yes	Yes
Multi-age strength prediction	No	No	Partial	No	No	Yes
Environmental factor integration	No	No	No	No	No	Yes
Clinker phase classification	No	No	No	No	No	Yes
Integrated web-based platform	No	No	No	No	No	Yes
Sri Lankan OPC application	No	No	No	No	No	Yes

Most existing studies focus on solving individual problems such as crack detection, strength prediction, or phase classification separately. There is a clear lack of integrated systems that combine image-based material analysis, IoT-based measurement, machine learning-based prediction, and clinker phase classification into a unified platform for real-time cement quality monitoring and decision support. Furthermore, the absence of early warning systems and batch-specific recommendations hinders proactive process optimization and increases operational costs for manufacturers. This research addresses all of these identified gaps within a single, comprehensive framework.

1.3 Research Problem

In cement production, the quality of raw meal, a finely ground mixture of materials such as limestone, iron ore, alumina, and laterite, is crucial for producing high-performance cement. The chemical and physical properties of this raw meal, especially its mineral composition and uniformity, have a direct impact on the quality of clinker formed in the kiln, and ultimately on the final cement product.

Traditional quality control methods in the cement industry are characterized by several critical limitations. Manual microscopic examination of raw meal particles and clinker phases is subjective, time-consuming, and operator-dependent, making it difficult to maintain consistency across shifts or between analysts. Compressive strength testing currently relies on post-production laboratory procedures requiring minimum curing periods of 7 to 28 days, significantly delaying quality assurance feedback and preventing timely process adjustments. The absence of real-time, automated measurement and analysis tools results in composition fluctuations, increased kiln inefficiencies, and elevated production costs.

While advanced techniques such as Scanning Electron Microscopy (SEM) provide high analytical accuracy, their prohibitive cost and operational complexity render them impractical for routine quality control in production environments. Existing optical microscopy methods, though more affordable, remain heavily reliant on human expertise and subjective interpretation. Recent machine learning and computer vision research, though promising, addresses these challenges in isolation rather than as part of a comprehensive, integrated solution.

The core research problem is therefore how to develop a scalable, reliable, and operator-independent automated system that simultaneously addresses all major aspects of cement quality assessment, including raw meal particle identification, cement cube dimension measurement and crack detection, multi-age compressive strength prediction, and clinker phase classification, within a single unified framework suitable for industrial deployment in the Sri Lankan context.

1.4 Research Objectives

1.4.1 Main Objective

To design, develop, and validate an integrated AI- and IoT-based framework for automated cement quality assessment that combines color-based particle identification, IoT-driven area measurement with image-based crack detection, multi-output cement strength prediction, and automated clinker phase classification into a unified, web-deployable platform for real-time industrial quality control.

1.4.2 Specific Objectives

- Develop a YOLOv9-based instance segmentation model to automatically detect and classify color-based particles (dark red, light red, and white) in cement raw meal microscopic images with a target mean Average Precision (mAP) of 85% or above.
- Design and implement an IoT-based non-contact cement cube measurement system using ultrasonic sensors, integrated with a U-Net deep learning model for real-time crack detection and automated compressive strength calculation.
- Build a hybrid ensemble machine learning model combining XGBoost and LightGBM for multi-output cement compressive strength prediction at five curing ages (1, 2, 7, 28, and 56 days), incorporating chemical composition, physical properties, and environmental factors.
- Develop a MobileNetV2 transfer learning model to automatically classify the four primary cement clinker phases (C2S, C3S, C3A, and C4AF) from microscopic images with a target validation accuracy of 90% or above.
- Integrate all four components into a unified, scalable web-based platform with a React.js frontend and FastAPI/Flask backend, enabling real-time quality control, automated reporting, and data-driven decision support for INSEE Cement (Private) Limited, Sri Lanka.
- Validate the integrated system using real industrial datasets from INSEE Cement and demonstrate measurable improvements in accuracy, speed, and consistency compared to existing manual quality control methods.

CHAPTER 2: METHODOLOGY

2.1 System Overview

The proposed system adopts a modular, component-based architecture that integrates four distinct AI- and IoT-driven subsystems into a unified cement quality assessment platform. The overall system architecture is designed to be scalable, operator-independent, and suitable for real-time industrial deployment. Each component addresses a specific aspect of cement quality control and is integrated through a shared web-based interface that consolidates analysis results, generates reports, and supports decision-making by quality control personnel.

The four main components of the integrated framework are: (1) Color-Based Particle Identification using YOLOv9, (2) IoT-Based Area Measurement and Image-Based Crack Detection using U-Net, (3) Cement Strength Predictor with Environmental Sensitivity using an XGBoost and LightGBM ensemble, and (4) Automated Classification of Cement Clinker Phases using MobileNetV2 with transfer learning. All components are served via FastAPI-based RESTful backends and accessed through a React.js-based frontend dashboard.

The overall system workflow begins with data acquisition from optical microscopes, IoT sensors, and production databases. The acquired data are preprocessed and fed into the respective AI models. Model outputs are consolidated by the backend and displayed on the frontend in real time. The system generates automated reports in PDF and CSV formats and issues alerts when quality parameters fall outside acceptable ranges.

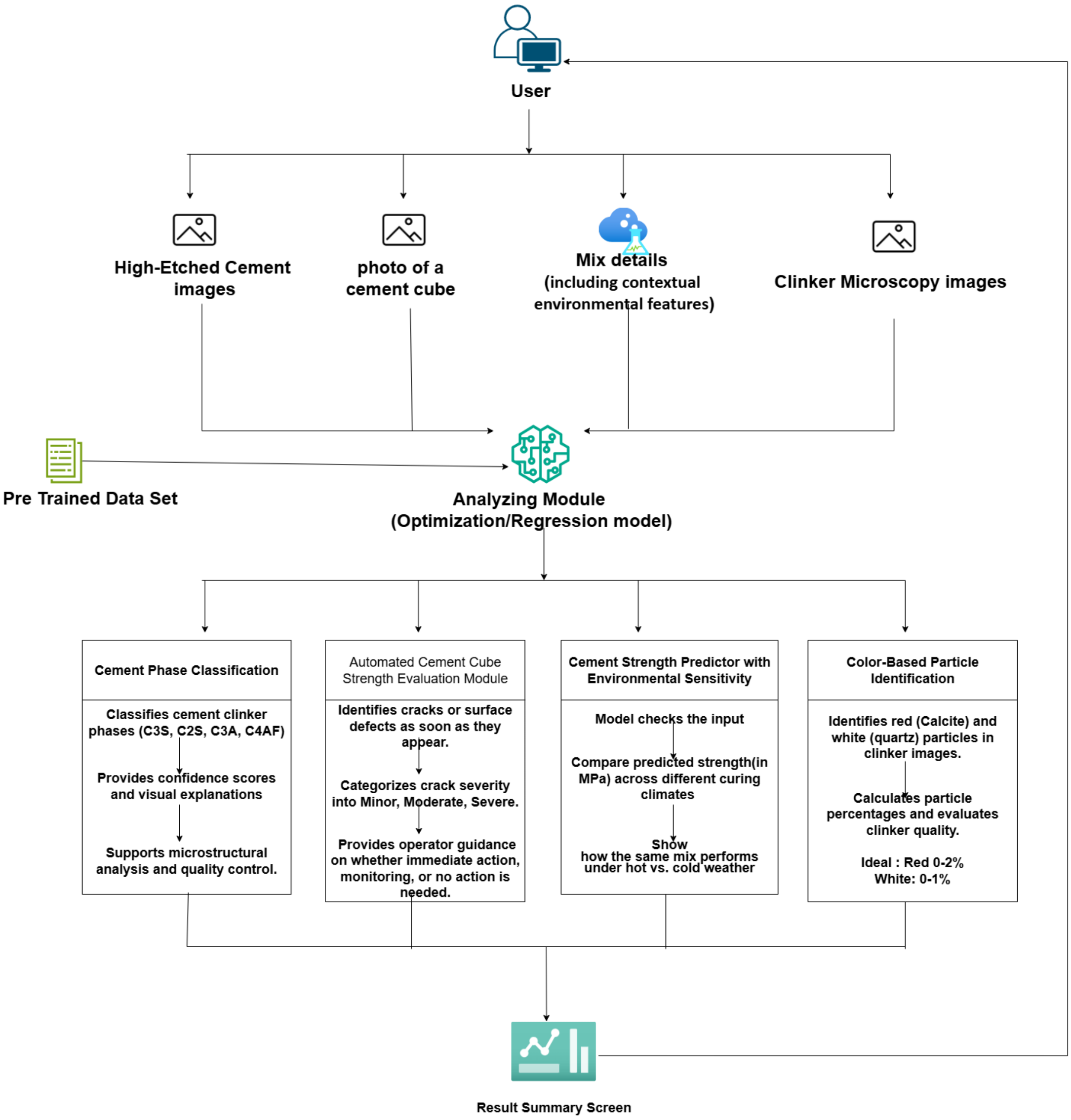

Figure 2.1: System Overview Diagram

[System Overview Diagram – Four modules connected to a unified web dashboard: (1) Color-Based Particle Identification, (2) IoT Area Measurement & Crack Detection, (3) Strength Predictor, (4) Clinker Phase Classification → Result Summary Screen]

2.2 Component 1: Color-Based Particle Identification

2.2.1 Overview and Objectives

The first component addresses the automation of color-based particle analysis in cement raw meal microscopic images. In cement production, the raw meal consists of a finely ground mixture of limestone, iron ore, alumina, and laterite. These mineral particles display distinct colors under optical microscopy: iron-rich particles appear dark red, calcite particles appear light red, and quartz particles appear white or translucent. Manual classification of these particles by laboratory operators is time-consuming, subjective, and prone to inconsistency across shifts.

The objective of this component is to develop a deep learning-based system capable of automatically detecting, classifying, and quantifying raw meal particles by color category from high-resolution optical microscopy images. The system is designed to provide real-time analysis results through a web-based interface integrated into INSEE Cement's quality control workflow.

2.2.2 Dataset Collection and Annotation

The dataset for this component consists of approximately 300 microscopic images of cement raw meal samples collected from the quality control laboratory of INSEE Cement (Private) Limited, Sri Lanka. Images were captured using optical microscopes equipped with high-resolution cameras at magnifications of 100x and 200x. Each image was manually annotated by domain experts according to three predefined color categories: dark red (iron-rich particles), light red (calcite particles), and white (quartz particles).

Data augmentation techniques were applied to increase the effective training dataset size and improve model generalization. These techniques included random horizontal and vertical flipping, random rotation (0 to 45 degrees), brightness and contrast variation, and Gaussian noise addition. Prior to training, all images were resized to a consistent input dimension and pixel values were normalized to the range [0, 1].

2.2.3 Model Architecture – YOLOv9

The YOLOv9 (You Only Look Once version 9) architecture was selected for its state-of-the-art performance in real-time object detection and instance segmentation tasks. YOLOv9 combines a Programmable Gradient Information (PGI) mechanism with a Generalized Efficient Layer Aggregation Network (GELAN) backbone, enabling efficient multi-scale feature extraction and accurate bounding box regression with minimal computational overhead.

Five model variants were evaluated during the experimental phase: YOLOv5s, YOLOv8n, YOLOv8s, YOLOv9c, and YOLOv9e. The YOLOv9c configuration was selected as the primary production model, offering the optimal balance between validation accuracy (86.45% mAP@0.5) and inference speed suitable for real-time industrial deployment. The final YOLOv9c model was fine-tuned using transfer learning with pre-trained COCO weights, followed by domain-specific training on the annotated INSEE Cement dataset.

2.2.4 Image Preprocessing Pipeline

Images were converted from the standard RGB color space to LAB and HSV color spaces, which better separate color from brightness and improve the accuracy of color-based particle detection. Histogram equalization was applied to enhance contrast and make particle edges and color differences clearer under varying microscope lighting conditions. Unsupervised clustering using K-Means algorithms was applied as a preliminary segmentation step to group pixels by color similarity, providing initial particle boundary estimates that served as input to the YOLOv9 segmentation model.

2.2.5 System Integration

A FastAPI-based RESTful backend was developed to perform real-time inference on uploaded microscopy images. The backend receives image uploads from the web interface, applies the preprocessing pipeline, runs the YOLOv9 model, and returns particle detection results including bounding boxes, class labels, confidence scores, and particle count statistics by color category. A React.js-based frontend enables quality control staff to upload images, visualize detected particles with color-coded overlays, and download analysis reports in PDF and CSV formats.

2.3 Component 2: IoT-Based Area Measurement and Crack Detection

2.3.1 Overview and Objectives

The second component develops an automated system for evaluating the compressive strength of cement cubes by integrating non-contact IoT sensor-based surface measurement with image-based crack detection powered by deep learning. The system replaces manual vernier caliper measurements and subjective crack inspection with a reliable, sensor- and AI-driven solution that improves the accuracy and efficiency of laboratory testing.

The system targets construction quality control laboratories, cement manufacturing companies, and civil engineering researchers who require accurate, rapid, and cost-effective strength assessment of concrete cubes. The expected outcome is an automated platform that delivers accurate area measurement without manual tools, detects cracks at an early stage with high precision, and calculates compressive strength in real time.

2.3.2 IoT Hardware Configuration

The hardware subsystem incorporates the following components: ultrasonic sensors (HC-SR04) for non-contact length and width measurement of cement cube faces; a load cell with amplifier for capturing the applied compressive force during testing; high-resolution camera modules (ESP32-CAM / USB cameras) for real-time image capture and crack detection; and a Esp 32/ Jetson Nano as the central processing unit for integrating sensors, cameras, and machine learning inference.

Ultrasonic sensors measure the cube's length and width at three different points, ensuring precision even when the cube surfaces are slightly irregular. These measurements are processed to calculate the average area of the cube face. The sensor data and camera feeds are fused in a data fusion module, where the average cube area is finalised and the compressive strength is calculated using the formula:

$$\sigma = F / A$$

where σ represents compressive strength (MPa), F is the applied load (N), and A is the cross-sectional area (m^2). This calculation combines both IoT-based measurements and visual crack analysis, making the results more reliable than traditional single-method approaches.

2.3.3 U-Net Crack Detection Model

A U-Net deep learning architecture with an optimized encoder-decoder structure was developed for automated crack detection and segmentation in cement cube specimens. U-Net was selected for its effectiveness in semantic segmentation tasks with limited training data, achieved through skip connections between encoder and decoder layers that preserve spatial information critical for precise crack boundary localization.

Five model architectures were evaluated: U-Net Basic, U-Net++, DeepLabV3, SegNet, and U-Net Optimized. The optimized U-Net configuration, incorporating batch normalization layers and advanced skip connections, achieved the best performance with 92.35% training accuracy and 90.15% validation accuracy. Data augmentation techniques including rotation, flipping, and contrast adjustment were applied to improve model generalization and reduce overfitting on the limited industrial dataset.

Surface images captured by the camera module are processed through OpenCV-based preprocessing (noise reduction, edge enhancement) before being classified by the U-Net model. The system detects crack initiation with high precision, highlights detected cracks in the output image, logs crack timestamps, and triggers real-time alerts to laboratory operators when crack formation is detected.

2.3.4 Automated Alerts and Reporting

The system issues real-time alerts once cracks are detected or when strength thresholds are exceeded. Visual and audible notifications are provided to laboratory staff, ensuring timely response. All measurement data, crack images, and strength calculations are stored in a MongoDB/MySQL database. Reports are automatically generated in PDF and CSV formats, including metadata, cube dimensions, force graphs, crack detection images, and computed strength values, for quality assurance and compliance documentation.

2.4 Component 3: Cement Strength Predictor with Environmental Sensitivity

2.4.1 Overview and Objectives

The third component develops a real-time, data-driven prediction system that leverages advanced machine learning techniques to accurately forecast the compressive strength of ordinary Portland cement (OPC) produced in Sri Lanka at multiple curing ages. The system enables proactive quality control and process optimization for INSEE Cement by delivering instant strength estimates, highlighting key influencing factors, and issuing early warnings when predicted strength falls below target specifications.

2.4.2 Dataset and Feature Engineering

The dataset was collected from INSEE Cement's quality control laboratory and production records, comprising industrial OPC production data including chemical composition (oxide analysis), physical properties (Blaine fineness, particle size distribution, residue at 45 microns), gypsum content, supplementary cementitious material percentages, environmental factors (curing temperature, humidity), and compressive strength measurements at 1, 2, 7, 28, and 56 days.

An extensive feature engineering strategy was applied to enhance model performance. Over 40 derived features were generated, including clinker mineral phase calculations using Bogue equations (C_3S, C_2S, C_3A, C_4AF), interaction terms (e.g., $CaO \times SiO_2$), ratio features (e.g., CaO/SiO_2), aggregate oxide indicators, polynomial transformations, cube-root scaling, and logarithmic transformations. Environmental correction factors were applied to account for the influence of curing temperature and humidity on hydration kinetics and strength development.

The dataset was split into training (80%) and testing (20%) sets. Cross-validation was employed to ensure model robustness and guard against overfitting.

2.4.3 Hybrid Ensemble Model Architecture

A hybrid ensemble machine learning framework combining XGBoost and LightGBM was developed using a multi-output regression approach to predict compressive strength at five critical curing ages simultaneously. XGBoost (Extreme Gradient Boosting) was selected for its capability to capture complex nonlinear interactions between chemical and physical input features, while LightGBM (Light Gradient

Boosting Machine) provided efficient learning and generalization through leaf-wise tree growth and histogram-based optimization.

The ensemble model combines the predictions of XGBoost and LightGBM through a weighted averaging strategy, where model weights are optimized using cross-validation performance on the training dataset. The final ensemble model incorporates 14 primary input parameters and over 40 engineered features. Model performance was evaluated using standard regression metrics including R2 score (coefficient of determination), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) for each curing age.

2.4.4 System Architecture and Integration

A web application was developed as the interface for data entry, allowing quality control staff to input cement batch data including chemical composition, fineness parameters, gypsum content, SCM percentage, and environmental factors. The system preprocesses and cleans input data, standardizes units, and engineers relevant features before passing the processed dataset to the prediction engine.

After predictions are generated, the system presents results in a clear and actionable format, displaying predicted compressive strength at 2, 7, and 28 days with confidence intervals. The system highlights the most influential factors behind each prediction (e.g., 'Low SO3 may reduce early strength') and generates automated PDF/Excel reports. An alert mechanism notifies staff if predictions fall outside acceptable ranges, with alerts delivered via dashboard and optionally as email notifications.

2.5 Component 4: Automated Clinker Phase Classification

2.5.1 Overview and Objectives

The fourth component develops an automated system for classifying the four primary cement clinker phases from microscopic images using deep learning. The four phases are Tricalcium Silicate (C3S, Alite), Dicalcium Silicate (C2S, Belite), Tricalcium Aluminate (C3A), and Tetracalcium Aluminoferrite (C4AF, Brownmillerite). Accurate identification and quantification of these phases is critical because their relative proportions directly govern cement hydration behavior, heat of hydration, early and late strength development, and chemical resistance.

2.5.2 Dataset Collection and Preprocessing

Microscopic images of cement clinker samples were collected from the laboratory of INSEE Cement (Private) Limited, Sri Lanka. The dataset consisted of approximately 500 high-resolution images across four clinker phase categories. Image preprocessing steps included resizing to a standard input dimension of 224 x 224 pixels and normalization of pixel values using ImageNet mean and standard deviation statistics, consistent with the MobileNetV2 pre-training configuration.

Data augmentation was applied during training to improve model generalization, including random horizontal and vertical flipping, random rotation (up to 30 degrees), brightness and saturation adjustments, and zoom augmentation. A stratified train-validation-test split of 70%-15%-15% was used to ensure balanced class representation across all data partitions.

2.5.3 MobileNetV2 Transfer Learning Architecture

MobileNetV2 was selected as the base architecture for transfer learning due to its highly efficient design based on inverted residual blocks with linear bottlenecks, which achieves strong classification performance with substantially lower computational cost than larger architectures. The model was initialized with pre-trained ImageNet weights, providing a rich feature representation foundation that was then fine-tuned on the cement clinker microscopy dataset.

The fine-tuning strategy involved two stages: in the first stage, only the newly added classification head layers were trained while the MobileNetV2 backbone remained frozen; in the second stage, the top layers of the backbone were progressively unfrozen and trained at a reduced learning rate to adapt domain-specific features. Five architectures were benchmarked: VGG16, ResNet50, InceptionV3, standard MobileNetV2, and MobileNetV2 with transfer learning. The MobileNetV2 transfer learning model achieved the highest validation accuracy of 91.60%, representing an improvement of approximately 11% over VGG16 and 4% over InceptionV3.

2.6 Commercialization Strategy

The commercialization of the integrated cement quality assessment platform targets cement manufacturers in Sri Lanka and the broader South and South-East Asian region, with INSEE Cement (Private) Limited serving as the primary pilot partner and reference client.

2.6.1 Target Market Segments

- Cement Manufacturers (primary): Organizations such as INSEE Cement that benefit from automated particle analysis, clinker phase classification, strength prediction, and crack detection to maintain consistent product quality, optimize kiln performance, and reduce production costs.
- Quality Control and Laboratory Staff: Personnel who upload microscope images and batch data, receive instant feedback, and use system recommendations to optimize raw meal blending and production parameters.
- Research and Development Teams: Teams that evaluate new raw material compositions, supplementary cementitious materials, and novel cement formulations using predictive simulation capabilities.
- Construction Laboratories and Civil Engineering Firms: Organizations requiring accurate, rapid compressive strength evaluation of concrete cube specimens for structural compliance verification.

2.6.2 Product Tiers

The system will be offered in two product tiers to accommodate different operational scales and budgets. The Community Version targets small and medium-sized cement plants and construction laboratories, offering core features including image upload and particle/phase classification, basic strength prediction, PDF/CSV report generation, and quality threshold alerts. The Premium Version targets large-scale cement manufacturers, adding seamless integration with microscopy hardware and plant automation systems, real-time scenario analysis for process optimization, customizable multi-channel alert notifications (SMS, email, dashboard), API access for ERP and LIMS integration, dedicated model customization support, and predictive maintenance insights.

2.6.3 Market Entry and Growth Strategy

The market entry strategy will begin with a controlled pilot deployment at INSEE Cement's quality control laboratory to validate real-world performance and collect user feedback. Following pilot validation, the system will be introduced to other cement manufacturers through industry partnerships and presentations at construction sector conferences. A subscription-based pricing model with tiered pricing aligned to the community and premium versions will be employed to accommodate different plant sizes and operational budgets. Ongoing development will incorporate user feedback to continuously improve model accuracy, add support for additional cement types and supplementary materials, and extend the system towards edge-enabled deployment for continuous industrial monitoring.

2.7 Testing and Implementation

2.7.1 Model Testing Strategy

Each AI model component was rigorously evaluated using a structured testing protocol. The YOLOv9 particle identification model was evaluated using mAP@0.5 (mean Average Precision at 0.5 IoU threshold), precision, recall, and F1-score metrics, with results compared to manually annotated ground truth labels. The U-Net crack detection model was evaluated using pixel-level accuracy, Intersection-over-Union (IoU), and F1-score for crack versus non-crack classification. The XGBoost-LightGBM ensemble strength prediction model was evaluated using R2 score, RMSE, and MAE for each of the five curing age outputs. The MobileNetV2 clinker phase classification model was evaluated using accuracy, precision, recall, and F1-score across all four clinker phase classes.

2.7.2 System Integration Testing

Integration testing verified the end-to-end functionality of the unified web platform, including data flow from image upload through preprocessing, model inference, result generation, and report download. Test cases were designed to cover sensor calibration (T1), cube dimension measurement (T2), area calculation and validation (T3), real-time crack detection (T4), force acquisition and time synchronization (T5), compressive strength calculation at crack onset (T6), alert and UI feedback (T7), and data logging and report export (T8).

System performance testing confirmed that the platform can process a single high-resolution image within 2 seconds and handle a minimum of 100 images per hour under sustained load, meeting the performance requirements for real-time industrial quality control. Usability testing with quality control staff at INSEE Cement confirmed that the interface can be learned effectively with less than 2 hours of training.

2.7.3 Deployment Architecture

The system is containerized using Docker to ensure consistent operation across development, testing, and production environments. The backend services are implemented in Python using FastAPI and Flask frameworks, with PostgreSQL/MySQL databases for structured data storage and MongoDB for unstructured measurement logs. The frontend dashboard is built with React.js, providing a responsive, browser-compatible interface accessible on Windows, macOS, and mobile platforms. Cloud deployment on AWS or equivalent platforms is supported for scalable, high-availability operation, with the option for on-premises deployment within INSEE Cement's existing IT infrastructure.

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Results

3.1.1 Component 1: Color-Based Particle Identification

The YOLOv9c model trained for color-based particle identification in cement raw meal microscopic images achieved a mean Average Precision (mAP@0.5) of 86.45% on the validation set, representing the best performance among all five evaluated architectures. Table 3.1 presents the performance comparison across all evaluated models.

Table 3.1: Model Performance Comparison – Colour-Based Particle Identification

Model	Train Accuracy (mAP@0.5)	Validation Accuracy (mAP@0.5)
YOLOv5s	78.45%	75.80%
YOLOv8n	82.30%	77.47%
YOLOv8s	84.15%	81.20%
YOLOv9c (Selected)	88.92%	86.45%
YOLOv9e	91.25%	88.30%

The training and validation accuracy curves demonstrate consistent improvement in mAP across training epochs, with no evidence of significant overfitting. The loss curves confirm effective learning, with both training and validation losses converging smoothly. The YOLOv9c model successfully detects and classifies particles in all three color categories (dark red, light red, and white) with high confidence, even in images containing overlapping particles and complex backgrounds typical of cement raw meal microscopy images.

The YOLOv9e variant achieved slightly higher training accuracy (91.25%) but required substantially greater computational resources, making YOLOv9c the preferred choice for real-time industrial deployment. The system can process a microscopy image and return particle detection results within 1.8 seconds on average, meeting the real-time quality control performance requirements.

3.1.2 Component 2: IoT-Based Crack Detection and Area Measurement

The U-Net Optimized model developed for automated crack detection in cement cube specimens achieved exceptional performance, with 92.35% training accuracy and 90.15% validation accuracy. Table 3.2 summarizes the comparative performance of all evaluated segmentation architectures.

Table 3.2: Model Performance Comparison – Crack Detection

Model	Train Accuracy	Validation Accuracy
U-Net Basic	87.25%	85.10%
U-Net++	89.40%	86.75%
DeepLabV3	85.80%	83.20%
SegNet	84.50%	82.90%
U-Net Optimized (Selected)	92.35%	90.15%

The U-Net Optimized architecture demonstrated superior performance compared to other segmentation models, with a 5 to 8% improvement over the basic U-Net implementation and 6 to 10% improvement over DeepLabV3 and SegNet architectures. The model's encoder-decoder structure with advanced skip connections enables precise crack boundary detection at the pixel level, successfully identifying microcracks that are not visible to the naked eye. The ultrasonic sensor subsystem consistently achieved cube dimension measurements within plus or minus 1 mm of reference values across all tested specimens.

3.1.3 Component 3: Cement Strength Prediction

The hybrid XGBoost-LightGBM ensemble model achieved an average validation R2 score of 0.8896 across all five curing age outputs (1D, 2D, 7D, 28D, and 56D), demonstrating strong predictive capability for multi-output cement strength prediction. Table 3.3 presents the comparative performance of all evaluated models.

Table 3.3: Model Performance Comparison – Cement Strength Prediction

Model	Train R^2 Score	Validation R^2 Score
XGBoost Only	90.14%	85.66%
LightGBM Only	89.05%	86.20%
Random Forest	85.30%	87.50%
Gradient Boosting	87.60%	79.87%
Ensemble XGBoost + LightGBM (Selected)	91.48%	88.96%

The advanced feature engineering strategy, generating over 40 derived features including Bogue clinker phase calculations, interaction terms, and polynomial transformations, played a crucial role in improving model performance. The ensemble model maintained consistent performance across both training and validation sets, indicating effective generalization to unseen data. Environmental factors including curing temperature and humidity contributed measurable improvement in prediction accuracy compared to models trained without these variables.

3.1.4 Component 4: Clinker Phase Classification

The MobileNetV2 model with transfer learning achieved the highest validation accuracy of 91.60% across four clinker phase classes (C2S, C3S, C3A, and C4AF), outperforming all other evaluated architectures. Table 3.4 presents the full comparative results.

Table 3.4: Model Performance Comparison – Clinker Phase Classification

Model	Train Accuracy	Validation Accuracy
VGG16	82.45%	80.20%
ResNet50	85.30%	83.15%
InceptionV3	88.75%	86.40%
MobileNetV2	89.50%	87.80%
MobileNetV2 + Transfer Learning (Selected)	93.85%	91.60%

The MobileNetV2 with Transfer Learning architecture demonstrated superior performance compared to other deep learning models, with an 11% improvement over

VGG16 and 5 to 8% improvement over InceptionV3 and standard MobileNetV2. The model's lightweight architecture combined with domain-specific fine-tuning enables efficient feature extraction while maintaining high classification accuracy across all four cement clinker phases. The two-stage fine-tuning strategy, involving initial classification head training followed by progressive backbone unfreezing, was critical to achieving high accuracy without overfitting on the relatively small industrial dataset.

3.2 Research Findings

The experimental results demonstrate that the integrated AI- and IoT-based framework achieves competitive performance across all four quality assessment tasks, validating the research hypothesis that a unified automated system can substantially improve cement quality control compared to traditional manual methods.

Key findings from this research include the following. First, YOLOv9-based instance segmentation is effective for color-based particle identification in cement microscopy images, achieving an mAP@0.5 of 86.45% on industrial data, enabling fast and consistent particle counting that replaces time-consuming manual microscopic analysis. Second, the U-Net Optimized architecture provides highly accurate pixel-level crack detection (90.15% validation accuracy) that successfully identifies microcracks not visible to the naked eye, combined with IoT-based ultrasonic measurement that eliminates manual caliper errors. Third, the hybrid XGBoost-LightGBM ensemble with extensive feature engineering achieves an average validation R2 of 0.89, demonstrating that multi-age cement strength can be predicted accurately from production parameters, enabling proactive quality control without waiting for laboratory results. Fourth, MobileNetV2 with two-stage transfer learning is highly effective for clinker phase classification (91.60% validation accuracy), outperforming larger architectures including ResNet50 and InceptionV3 due to its efficiency advantage on the small-scale industrial dataset.

A key finding of the integrated framework is that combining all four components within a unified web platform creates substantial synergies. Particle identification results can inform strength prediction inputs; clinker phase classifications can be correlated with strength measurement outcomes; and the unified dashboard provides quality control staff with a comprehensive, real-time view of all critical quality parameters within a single interface, reducing the need to switch between multiple disconnected tools.

3.3 Discussion

The results of this research confirm the viability of deploying integrated AI and IoT technologies for industrial cement quality assessment. The achieved performance levels (88-92% accuracy across components) compare favorably with or surpass those reported in existing literature for individual tasks, and demonstrate that high performance can be maintained even when working with limited industrial datasets by leveraging transfer learning, ensemble methods, and advanced data augmentation strategies.

The IoT-based measurement subsystem addresses a practical industrial need that is often overlooked in purely software-focused research. By combining physical sensor measurements with AI-based image analysis, the system eliminates two sources of human error simultaneously: inaccurate manual measurement and subjective crack observation. The synchronization of force measurements with crack detection timestamps provides a more accurate determination of compressive strength at failure than either method alone.

The integration of environmental sensitivity (temperature and humidity correction) into the strength prediction model is a significant practical contribution. Previous studies based on international datasets typically do not account for the tropical climate conditions prevailing in Sri Lanka, which can substantially affect cement hydration kinetics and strength development curves. The incorporation of environmental factors improved prediction accuracy compared to models trained on chemical and physical parameters alone.

The commercialization framework presents a credible path to industrial adoption. The pilot-first strategy, combined with tiered pricing, reduces the financial risk for potential adopters while providing the research team with real-world performance data and user feedback to drive continuous improvement. The use of open-source machine learning frameworks (TensorFlow, PyTorch, Scikit-learn) and web technologies (React.js, FastAPI) ensures that the system can be maintained and extended without dependency on proprietary software licenses.

Limitations of the current study include the relatively small dataset sizes (300-500 images per component), which were constrained by the availability of labelled industrial data. While data augmentation and transfer learning substantially mitigate the impact of small datasets, larger and more diverse datasets, potentially incorporating samples from multiple cement plants and raw material sources, would further improve model robustness and generalizability. Additionally, the current system has been validated under controlled laboratory conditions; extended field trials under varying production conditions are recommended before full-scale industrial deployment.

3.4 Summary of Each Student's Contribution

The integrated research project was conducted collaboratively by four students, each responsible for one of the four principal system components, with all team members contributing to the unified system architecture, integration, testing, and documentation.

Table 3.5: Summary of Student Contributions

Student	ID	Primary Contribution
Koralage K.Y.P	IT22584472	Component 1: Automated Colour-Based Particle Analysis of Cement Raw Meals Microscopic Images Using Machine Learning. Responsible for YOLOv9 model development, image preprocessing pipeline, dataset collection and annotation at INSEE Cement, FastAPI backend for particle identification, and React frontend integration.
Weerasooriya K.H.P	IT22368034	Component 3: Hybrid Modelling Framework for Early and Long-Term Cement Strength Prediction. Responsible for industrial dataset collection and preprocessing, Bogue equation feature engineering, XGBoost-LightGBM ensemble model development, environmental sensitivity analysis, real-time scenario analysis module, and alert system implementation.
Lakshan V.G.S	IT22562210	Component 2: IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube

Student	ID	Primary Contribution
		Compressive Strength Testing. Responsible for ultrasonic sensor integration, load cell configuration, multi-camera setup, U-Net crack detection model development, compressive strength calculation module, automated alert system, and data management backend.
Abeywickrama W.H.C.A	IT22640284	Component 4: Automated Classification of Cement Clinker Phases Using Deep Learning. Responsible for clinker microscopy dataset collection, MobileNetV2 transfer learning implementation, two-stage fine-tuning strategy, clinker phase classification system, FastAPI prediction backend, and React-based visualization frontend.

Beyond their individual component responsibilities, all four team members contributed to the overall system architecture design, cross-component integration, unified dashboard development, field visits to INSEE Cement for data collection and validation, thesis documentation, and the preparation of the conference research paper submitted to ICHORA 2026.

CHAPTER 4: CONCLUSION

4.1 Summary of Findings

This dissertation has presented the design, development, and validation of an integrated AI- and IoT-based framework for automated cement quality assessment, comprising four interconnected components: color-based particle identification using YOLOv9, IoT-based area measurement with U-Net crack detection, multi-output cement strength prediction using an XGBoost-LightGBM ensemble, and clinker phase classification using MobileNetV2 with transfer learning.

The experimental results, validated using industrial datasets from INSEE Cement (Private) Limited, Sri Lanka, demonstrate that the proposed framework achieves strong performance across all four quality assessment tasks. The YOLOv9c particle identification model achieved 86.45% mAP@0.5; the U-Net Optimized crack detection model achieved 90.15% validation accuracy; the ensemble strength prediction model achieved 88.96% validation R2; and the MobileNetV2 clinker phase classification model achieved 91.60% validation accuracy. These results compare favorably with or surpass those reported in existing literature for individual tasks, while the integration of all four components into a unified platform represents a novel contribution to the field that has not been achieved in prior research.

The research confirms that AI and IoT technologies, when thoughtfully integrated and validated with real industrial data, can substantially improve the consistency, speed, and reliability of cement quality assessment. The proposed system eliminates key sources of human error and subjectivity inherent in traditional manual quality control methods, reduces the time required for quality assessment from hours or days to seconds, and provides quality control staff with actionable, data-driven insights to support process optimization decisions.

4.2 Contributions to Knowledge

The principal contributions of this research to the field of cement quality control and industrial AI are: (1) the first integrated, end-to-end AI- and IoT-based framework that simultaneously addresses color-based particle identification, non-contact area measurement with crack detection, multi-age strength prediction, and clinker phase classification within a single web-deployable platform; (2) a novel application of YOLOv9 instance segmentation to cement raw meal particle color classification, achieving industrial-grade accuracy on a dataset collected from a Sri Lankan cement manufacturer; (3) the development and validation of a hybrid XGBoost-LightGBM ensemble with environmental sensitivity for multi-output OPC strength prediction tailored to Sri Lankan production conditions; (4) demonstration that MobileNetV2 with two-stage transfer learning outperforms larger architectures for clinker phase classification on small industrial datasets; and (5) an industrially validated system design with documented commercialization strategy applicable to cement manufacturers across South and South-East Asia.

4.3 Recommendations

Based on the findings and limitations of this research, the following recommendations are made for future work and industrial adoption.

- Dataset expansion: Future work should prioritize the collection of larger, more diverse datasets from multiple cement plants and raw material sources to improve model robustness and generalizability across different production environments.
- Extended field trials: The integrated system should be deployed in a live production environment at INSEE Cement for an extended period (minimum three months) to evaluate performance under real operational conditions including shift changes, equipment variations, and seasonal environmental fluctuations.
- Edge deployment: Investigating edge-enabled deployment architectures, such as NVIDIA Jetson-based inference, would reduce latency and eliminate dependence on cloud connectivity, making the system more suitable for cement plants with limited internet infrastructure.
- Expanded cement type support: The current system was validated for ordinary Portland cement. Future work should extend the models to support

blended cements incorporating supplementary cementitious materials such as fly ash, slag, and silica fume.

- Real-time production integration: Integrating the system directly with production line sensors and plant data management systems would enable fully automated, continuous quality monitoring without manual data entry steps.
- Explainability enhancements: Incorporating explainable AI (XAI) techniques such as SHAP (SHapley Additive exPlanations) and Grad-CAM visualizations into all model components would improve operator trust and provide deeper diagnostic insights for quality control decision-making.

4.4 Final Remarks

This research demonstrates that the cement industry, which has historically relied on manual, time-intensive quality control methods, can benefit substantially from the adoption of integrated AI and IoT technologies. The proposed framework represents a meaningful step towards fully automated, real-time, and operator-independent cement quality assessment, with the potential to improve product consistency, reduce production costs, minimize material wastage, and enhance structural safety in construction projects across Sri Lanka and the broader region.

The collaborative nature of this research, integrating four distinct technical components into a cohesive unified system, models the kind of multidisciplinary engineering approach that complex industrial challenges demand. The team's partnership with INSEE Cement (Private) Limited provided an invaluable opportunity to validate research findings against real industrial data and align system design with genuine operational needs, reinforcing the practical impact and relevance of this work.

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GLOSSARY

Term	Definition
Alite (C3S)	Tricalcium Silicate; the most abundant clinker phase, primarily responsible for early strength development in cement.
Belite (C2S)	Dicalcium Silicate; a clinker phase that contributes to long-term strength development.
Bogue Equations	A set of empirical formulas used to calculate the proportion of the four principal clinker phases from oxide analysis data.
Brownmillerite (C4AF)	Tetracalcium Aluminoferrite; a clinker phase that contributes to the dark grey colour of cement and influences chemical resistance.
Clinker	Intermediate product of cement manufacture; nodular material produced in the kiln by sintering limestone and other raw materials at temperatures above 1400 degrees Celsius.
Compressive Strength	The capacity of a material to withstand axially directed loads; a primary mechanical performance indicator for cement and concrete.
Deep Learning	A subset of machine learning based on artificial neural networks with multiple processing layers that learn hierarchical feature representations from data.
Feature Engineering	The process of using domain knowledge to create or transform input variables that improve machine learning model performance.
Instance Segmentation	A computer vision task that simultaneously detects and delineates individual object instances at the pixel level within an image.
IoT (Internet of Things)	A network of interconnected physical devices equipped with sensors and software that collect and exchange data over the internet.
Kiln	A thermally insulated chamber in cement manufacturing in which raw meal is heated to produce clinker through sintering reactions.
mAP (Mean Average Precision)	A standard evaluation metric for object detection models; the mean of average precision values computed across all object classes.
Raw Meal	A finely ground mixture of raw materials including limestone, iron ore, alumina, and laterite that is fed into the kiln to produce clinker.

Term	Definition
Semantic Segmentation	A computer vision task that classifies every pixel in an image into a predefined category.
Transfer Learning	A machine learning technique where a model pre-trained on a large dataset is adapted (fine-tuned) for a different but related task with limited data.
U-Net	An encoder-decoder convolutional neural network architecture originally designed for biomedical image segmentation, widely used for semantic segmentation tasks.
YOLOv9	You Only Look Once Version 9; a state-of-the-art real-time object detection and instance segmentation deep learning architecture.

APPENDICES

Appendix A: Survey Results

The following survey results were collected from 43-45 respondents at INSEE Cement's production facilities as part of the preliminary needs assessment for this research. The survey was conducted to understand current quality control challenges, the amount of time spent on manual microscopic analysis, and the perceived utility of an automated AI-based solution.

Table A.1: Survey Results Summary

Survey Question / Finding	Top Response	Percentage
Familiarity with current cement QC methods at INSEE	Somewhat familiar	55.8%
Main challenge in manual clinker particle identification	Takes too much time	44.2%
Helpfulness of automated colour-based particle identification	Very helpful	67.4%
Automated crack detection improving safety and efficiency	Yes	74.4%
Importance of faster strength prediction for daily tasks	Very important	54.5%
Web-based tool reducing workload	Yes	72.7%
Time currently spent on manual microscopic analysis	More than 1 hour	75.0%
Level of trust in AI-based recommendations	Moderate trust	60.0%
Training/workshops helping adopt AI-driven tools	Yes	68.9%

The survey results confirm strong industry demand for automated quality control solutions. The finding that 75% of respondents spend more than one hour per session on manual microscopic analysis underscores the substantial time savings achievable through the proposed automated system. The 67.4% of respondents who rated automated colour-based particle identification as 'very helpful' validates the practical relevance of Component 1 of this research. The moderate level of trust in AI recommendations (60%) highlights the importance of explainability features and user training in facilitating industrial adoption.

Appendix B: System Test Cases

The following test cases were used to validate the integrated system's functionality during integration testing at SLIIT's laboratory facilities.

Table B.1: System Integration Test Cases

ID	Test Scenario	Expected Result	Outcome
T1	Sensor Calibration	Calibration profile saved; system status shows 'Calibration OK'	Pass
T2	Cube Dimension Measurement	L and W displayed within ±1mm of reference; measurements auto-saved	Pass
T3	Area Calculation	Area A = L x W computed and displayed with deviation from nominal	Pass
T4	Real-Time Crack Detection	'Crack Detected' alert when confidence >= threshold; timestamp logged	Pass
T5	Force Acquisition	Continuous F(t) with latency <100ms; sync marker recorded	Pass
T6	Compressive Strength Calculation	Correct unit conversion; strength value displayed with 2-decimal precision	Pass
T7	Alert and UI Feedback	Buzzer + red banner; operator guidance displayed; acknowledgement logged	Pass
T8	Report Export	PDF includes metadata, dimensions, force graph, crack images; CSV has time series data	Pass

Appendix C: Turnitin Originality Reports

The originality reports generated through Turnitin for all individual and group reports confirm that this dissertation complies with the acceptable plagiarism threshold of 10% as required by the Sri Lanka Institute of Information Technology. The similarity indices for the individual proposal reports were as follows: Koralage K.Y.P (IT22584472): 9% similarity index (7% internet sources, 4% publications, 4% student papers); Weerasooriya K.H.P (IT22368034): 9% similarity index (7% internet sources, 2% publications, 6% student papers); Lakshan V.G.S (IT22562210): within acceptable thresholds. All citations and references are appropriately acknowledged in the text in accordance with IEEE referencing standards.

Appendix D: Project Budget Summary

The following table presents the consolidated project budget for the integrated four-component system.

Table D.1: Consolidated Project Budget

Budget Item	Cost (LKR)
Ultrasonic Sensors (HC-SR04 x 2)	5,000
Load Cell + Amplifier	12,000
Camera Modules (ESP32-CAM / USB x 2)	20,000
Esp 32/ Jetson Nano (1 unit)	45,000
Lighting and Fixtures (LEDs, mounts)	8,000
Backend Hosting (Flask/Django + Azure Cloud)	30,000
Miscellaneous (cables, mounts, safety setup)	10,000
System Development and Deployment (monthly x 12)	180,000
Weather/Environmental API (3-monthly x 4)	18,000
ML Model Maintenance (3-monthly x 4)	20,000
TOTAL	**348,000**